SYNTHETIC EXAMPLE - NOT A REAL TECHNOLOGY

Compact Radiator Panel for Small Satellites: Test Report

1. Background
Earlier studies showed that loop heat pipes can cool small spacecraft.
The project targets TRL 6 by the end of 2027.
This report covers the radiator panel component only.

2. Thermal vacuum test

A flight-like radiator panel was built from aluminium honeycomb with embedded heat pipes. The panel was tested in a thermal vacuum chamber from -40 C to +85 C over 12 cycles. Measured heat rejection was 41 W at 20 C, within 3 percent of the analytical prediction. All acceptance criteria in the test plan were met, with no leaks or degradation.

3. Next steps

The panel will be integrated with the full thermal control subsystem in 2027.
A flight demonstration on a 6U CubeSat is planned but not yet funded.